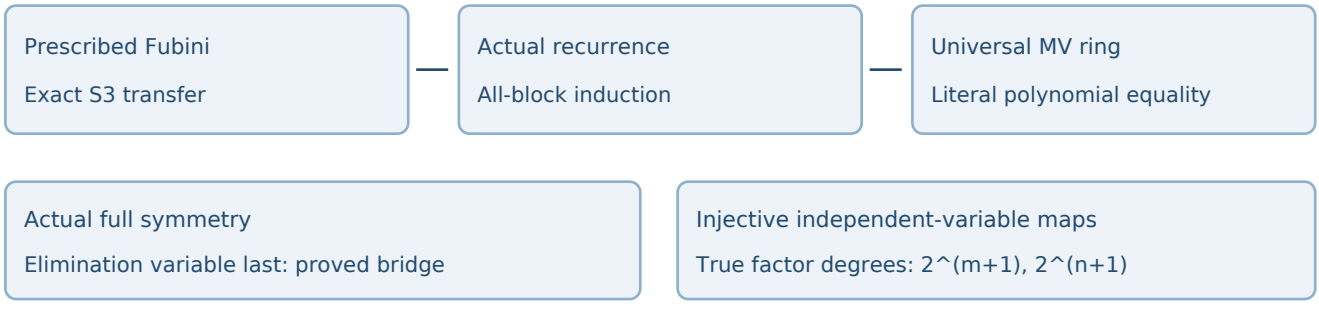
Semaev: every literal source-ordered split

Proved. For all fields and all m,n≥0, the actual source polynomials of arities m+3 and n+3, with elimination variable last, have natural degrees 2^(m+1) and 2^(n+1). Their resultant with these true dimensions equals the literal canonical polynomial of arity N=(m+2)+(n+2). Explicit Fin casts bind the variables. Every source index N≥4 and 1≤k≤N−3 is covered by k=n+1. Signed standing unverified. [pcc:unverified]



Proof. Independent generic leading coefficients and specialization prove fully prescribed Fubini. The actual S3 quadratic and proved even resultant sign give convolution transfer. Actual canonical recurrence, exact tuple/list bridges and full input symmetry prove the two-block induction. The target is the actual multivariate polynomial ring: universal evaluation and variable-block concatenation yield literal polynomial equality. This is not an inference from agreement at finite-field points. [pcc:unverified]

Degrees and source order. Coefficient evaluation at injectively embedded independent variables is identified with MVrename and proved injective. Mapping therefore preserves the actual canonical degrees. A separate ring-hom extensionality proof identifies the source kernel with partial evaluation of canonicalMv; full symmetry moves its free variable from first to last. No degree, ordering or split premise replaces these bridges. [pcc:unverified]

Specialization. The certificate also proves every two-block identity over any commutative domain under any coefficient map, allowing repeated coordinates and degree drops. Here prescribed dimensions are retained. The literal universal factors have true natural dimensions; a specialized factor can have a smaller natural degree. These are explicit distinct statements. [pcc:unverified]

Negative controls. General resultant interchange with reset natural degrees can give −1 on the left and 2 on the right, while prescribed dimensions agree. Independent b=(0,0,0,1) breaks S4 symmetry, but Lean proves these invariants cannot come from a rational Weierstrass curve. Both controls are actual inequalities in the closed certificate. Neither refutes the qualified recursion. [pcc:unverified]

Checks. Whole gate 82 declarations / 0 failures / 0 trusted native exceptions; Lean4.35.0 and Mathlib56a00585…. Complete declaration/import/config and artifact pins are in sources.json. Failed and successful source/log/exit records and the gate are retained losslessly. Only propext, Classical.choice and Quot.sound; no holes, custom axioms, native evaluator, raised limits, global elaboration flags or waivers. Exact PCC replay is separate; signed standing unverified. [pcc:unverified]

Sources and boundary. Semaev2004 Theorem1 equation2/proof, originally characteristic not2/3, and Kosters–Yeo2015 Section2/p3 general invariant model. This confirms the qualified recursion; no novelty or source refutation is claimed. The formal invariant-polynomial identity holds in all characteristics for actual coefficient models, without a new group interpretation for singular models. Higher-arity absolute irreducibility remains open, as do the other obligations in the all-fourteen map. KB unavailable; bounded primary/repository/library review recorded before edits. No new native experiment, paid compute or agent upload. [pcc:unverified]